

SAGI Summer School 2025: Introduction to Fourier Analysis

Sam Condon

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RC Circuit

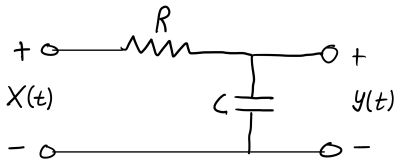
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RC Circuit

- ① We will start our journey in Fourier analysis with some motivation from circuit theory. We will predict the time domain output of a circuit given an arbitrary input.
- ② The circuit we will consider is the simple RC circuit shown below, with a voltage signal $x(t)$ applied to the input, and a voltage signal $y(t)$ on the output:



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- ❶ Applying Kirchoff's voltage law, with $I(t)$ being the current through the circuit:

$$x(t) = I(t)R + y(t) \quad (1)$$

- ❷ If $Q(t)$ is the charge on the capacitor plate, we also have:

$$y(t) = \frac{Q(t)}{C} \quad (2)$$

$$I(t) = \frac{dQ(t)}{dt} \quad (3)$$

- ❸ Substituting, we get a differential equation describing the circuit behavior:

$$x(t) = RC \frac{dy(t)}{dt} + y(t) \quad (4)$$

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- ➊ Now suppose that the input is sinusoidal $x(t) = e^{i\omega t} = \cos(\omega t) + i \sin(\omega t)$.
- ➋ Let's test as a possible output solution a scaled version of the input:
 $y(t) = Ae^{i\omega t}$
- ➌ Substituting these values into the differential equation:

$$e^{i\omega t} = RC \frac{d}{dt} (Ae^{i\omega t}) + Ae^{i\omega t} \quad (5)$$

$$= i\omega RCAe^{i\omega t} + Ae^{i\omega t} \quad (6)$$

$$= (1 + i\omega RC)Ae^{i\omega t} \quad (7)$$

- ➍ We see that:

$$y(t) = \left(\frac{1}{1 + i\omega RC} \right) x(t) = Ax(t) \quad (8)$$

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- ① Indeed, we see that the output is just a scaled version of the input!
- ② A fancy way to say this is that a complex exponential (sinusoid) is an **eigenfunction** of the differential operator describing the RC circuit.
- ③ Because the scaling of the input to the output A is frequency dependent, let's relabel:

$$A \rightarrow H(\omega) = \frac{1}{1 + i\omega RC} \quad (9)$$

so that:

$$y(t) = H(\omega)x(t) \quad (10)$$

- ④ Now, let's suppose that we take the input to be $x(t) = e^{i\omega_1 t} + e^{i\omega_2 t}$.
Motivated by the previous result, let's guess that:

$$y(t) = H(\omega_1)e^{i\omega_1 t} + H(\omega_2)e^{i\omega_2 t} \quad (11)$$

$$= \left(\frac{1}{1 + i\omega_1 RC} \right) e^{i\omega_1 t} + \left(\frac{1}{1 + i\omega_2 RC} \right) e^{i\omega_2 t} \quad (12)$$

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- 1 Let's check if this $y(t)$ is indeed a solution to eqn. (4).
- 2 Go back to the differential equation, substitute the input and the trial solution:

$$\begin{aligned} e^{i\omega_1 t} + e^{i\omega_2 t} &= \frac{i\omega_1 RC}{1 + i\omega_1 RC} e^{i\omega_1 t} + \frac{i\omega_2 RC}{1 + i\omega_2 RC} e^{i\omega_2 t} \\ &\quad + \frac{1}{1 + i\omega_1 RC} e^{i\omega_1 t} + \frac{1}{1 + i\omega_2 RC} e^{i\omega_2 t} \\ &= \left(\frac{1 + i\omega_1 RC}{1 + i\omega_1 RC} \right) e^{i\omega_1 t} + \left(\frac{1 + i\omega_2 RC}{1 + i\omega_2 RC} \right) e^{i\omega_2 t} \\ &= e^{i\omega_1 t} + e^{i\omega_2 t} \end{aligned}$$

- 3 Indeed, this $y(t)$ is a solution. Emboldened by this result, let's generalize to an arbitrary summation of complex exponentials.

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Statement

If the input signal to the RC circuit $x(t)$ is an arbitrary summation of complex exponentials:

$$x(t) = \sum_{n=0}^{\infty} a_n e^{i\omega_n t} \quad (13)$$

Then the output signal $y(t)$ of the circuit is:

$$y(t) = \sum_{n=0}^{\infty} H(\omega_n) a_n e^{i\omega_n t} \quad (14)$$

where:

$$H(\omega_n) = \frac{1}{1 + i\omega_n t} \quad (15)$$

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- ① This statement is actually quite general and applies to nearly every circuit that we will work with.
- ② It is really a statement that follows from linearity and time-invariance of the differential equation governing the circuit.
- ③ If we know the output of the circuit corresponding to a sinusoidal input with frequency ω , then we know the output corresponding to some arbitrary summation of sinusoids of various frequencies.
- ④ This is because complex exponentials (sinusoids) are **eigenfunctions** of **linear-time-invariant** operators.

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- ① In the real physical world, we take the real part of complex exponentials for our input, and we use the **magnitude** and **phase** of $H(\omega)$ to determine our output.
- ② That is, $x(t) = e^{i\omega t} \rightarrow \mathcal{R}\{e^{i\omega t}\} = \cos(\omega t)$ and:

$$y(t) = H(\omega)x(t) \rightarrow |H(\omega)| \cos(\omega t + \angle H(\omega)) \quad (16)$$

where:

$$|H(\omega)| = \sqrt{\mathcal{R}\{H(\omega)\}^2 + \mathcal{I}\{H(\omega)\}^2} \quad (17)$$

$$\angle H(\omega) = \tan^{-1} \left(\frac{\mathcal{I}\{H(\omega)\}}{\mathcal{R}\{H(\omega)\}} \right) \quad (18)$$

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- ❶ For the RC circuit, we have:

$$H(\omega) = \frac{1}{1 + i\omega RC} = \frac{1 - i\omega RC}{1 + (\omega RC)^2} \quad (19)$$

- ❷ So we see that the real and imaginary components are:

$$\mathcal{R}\{H(\omega)\} = \frac{1}{1 + (\omega RC)^2} \quad (20)$$

$$\mathcal{I}\{H(\omega)\} = -\frac{\omega RC}{1 + (\omega RC)^2} \quad (21)$$

- ❸ And:

$$|H(\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}, \quad \angle H(\omega) = \tan^{-1}(-\omega RC) \quad (22)$$

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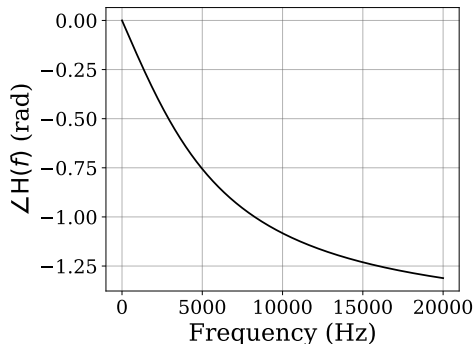
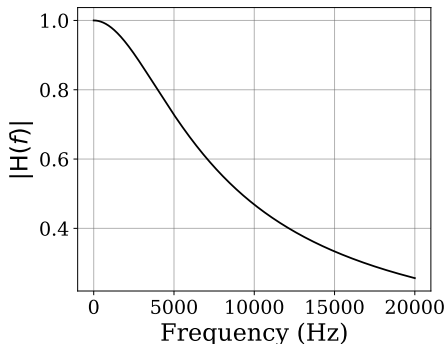
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- ❶ Let's make this a bit more concrete with a real circuit. let's take $R = 1 \text{ k}\Omega$ and $C = 30 \text{ nF}$.
- ❷ Recalling that we have $\omega = 2\pi f$, we can plot the **magnitude** and **phase** response of the circuit:



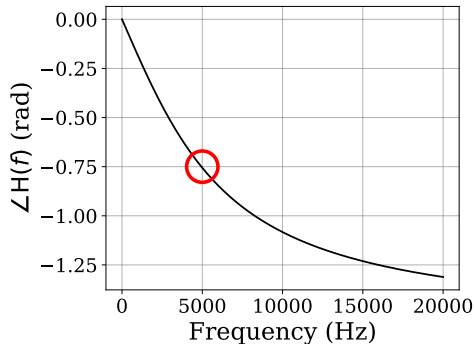
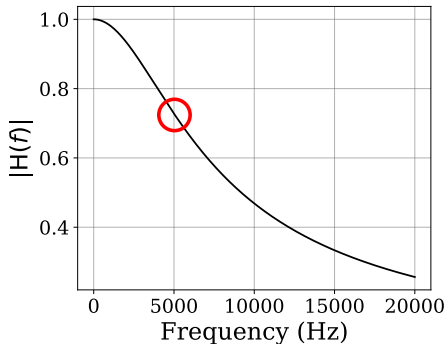
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- ① Taking $f = 5$ kHz, we can predict the output by finding the correct points on the magnitude and phase plots:



- ② We predict roughly:

$$y(t) = 0.7 \cdot \cos(2\pi ft - 0.75) \quad (23)$$

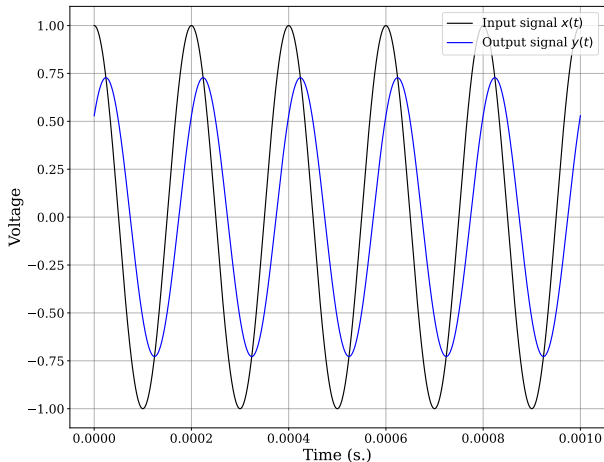
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① Plotting the input and predicted output together, we get:



② Let's compare this to a real circuit!

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attach pictures from the demonstration here

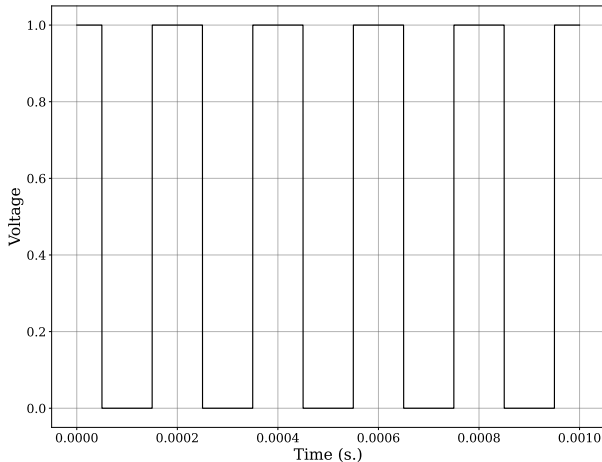
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- ① Now, let's try a slightly more complicated function to really test the bold statement of eqn. (14) above.
- ② Consider the square wave signal plotted below:



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- ① We will soon see that this signal can be represented as the following summation of sinusoids:

$$s(t) = \frac{1}{2} + 2 \sum_{n=0}^{\infty} \left[\frac{2 \sin(n\pi/2)}{n\pi} \right] \cos(2\pi fnt) \quad (24)$$

- ② Using equation (14) and (16) above, we can predict the output:

$$y(t) = \frac{|H(0)|}{2} \cos(\angle H(0)) \quad (25)$$

$$+ 2 \sum_{n=0}^{\infty} H(2\pi fnt) \left[\frac{2 \sin(n\pi/2)}{n\pi} \right] \cos(2\pi fnt + \angle H(2\pi fnt)) \quad (26)$$

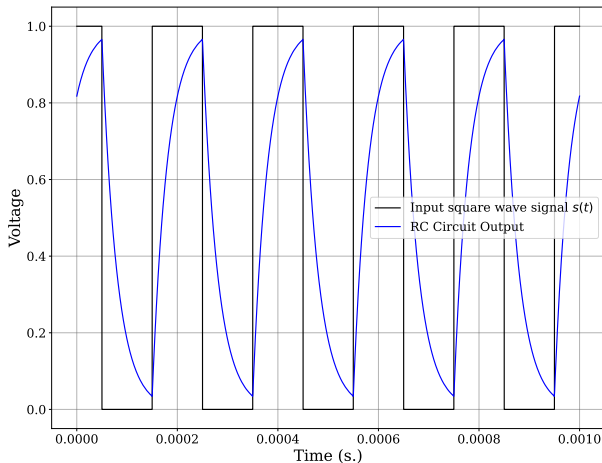
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- ① Using eqn. (25) and some python code, we predict the output of the circuit to be:



- ② Of course, we need to check this with an actual circuit!

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add pictures from the demonstration here

RC Circuit: Summary

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- ➊ In this lecture we analyzed an RC low-pass filter circuit. We saw that the output of the circuit to a sinusoidal input is just a scaled and phase shifted version of the input.
- ➋ We extended this result to say that if the input to the circuit is any summation of sinusoids, the output will be scaled and phase shifted versions of these same sinusoids.
- ➌ We checked both of these results by predicting the output of the filter to a single sinusoid and to a square wave and we verified that our predictions match the behavior of the real circuit.
- ➍ Fourier analysis consists of treating all input signals as summations of sinusoids and designing circuits and systems to manipulate these sinusoids in a desired manner.